

Logarithms

12.1 Introduction to logarithms

With the use of calculators firmly established, logarithmic tables are now rarely used for calculation. However, the theory of logarithms is important, for there are several scientific and engineering laws that involve the rules of logarithms.

If a number y can be written in the form a^x , then the index x is called the 'logarithm of y to the base of a ',

i.e.

$$\text{if } y = a^x \text{ then } x = \log_a y$$

Thus, since $1000 = 10^3$, then $3 = \log_{10} 1000$

Check this using the 'log' button on your calculator.

- (a) Logarithms having a base of 10 are called **common logarithms** and $\log_{10}$ is usually abbreviated to $\lg$. The following values may be checked by using a calculator:

$$\lg 17.9 = 1.2528 \dots,$$

$$\lg 462.7 = 2.6652 \dots$$

$$\text{and } \lg 0.0173 = -1.7619 \dots$$

- (b) Logarithms having a base of e (where ' e ' is a mathematical constant approximately equal to 2.7183) are called **hyperbolic, Napierian** or **natural logarithms**, and $\log_e$ is usually abbreviated to $\ln$.

The following values may be checked by using a calculator:

$$\ln 3.15 = 1.1474 \dots,$$

$$\ln 362.7 = 5.8935 \dots$$

$$\text{and } \ln 0.156 = -1.8578 \dots$$

For more on Napierian logarithms see Chapter 13.

12.2 Laws of logarithms

There are three laws of logarithms, which apply to any base:

- (i) To multiply two numbers:

$$\log (A \times B) = \log A + \log B$$

The following may be checked by using a calculator:

$$\lg 10 = 1,$$

$$\text{also } \lg 5 + \lg 2 = 0.69897 \dots$$

$$+ 0.301029 \dots = 1$$

$$\text{Hence } \lg(5 \times 2) = \lg 10 = \lg 5 + \lg 2$$

- (ii) To divide two numbers:

$$\log \left(\frac{A}{B} \right) = \log A - \log B$$

The following may be checked using a calculator:

$$\ln \left(\frac{5}{2} \right) = \ln 2.5 = 0.91629 \dots$$

$$\text{Also } \ln 5 - \ln 2 = 1.60943 \dots - 0.69314 \dots = 0.91629 \dots$$

$$\text{Hence } \ln \left(\frac{5}{2} \right) = \ln 5 - \ln 2$$

- (iii) To raise a number to a power:

$$\lg A^n = n \lg A$$

The following may be checked using a calculator:

$$\lg 5^2 = \lg 25 = 1.39794 \dots$$

Also $2 \lg 5 = 2 \times 0.69897 \dots = 1.39794 \dots$

Hence $\lg 5^2 = 2 \lg 5$

Problem 1. Evaluate

- (a) $\log_3 9$ (b) $\log_{10} 10$ (c) $\log_{16} 8$

- (a) Let $x = \log_3 9$ then $3^x = 9$ from the definition of a logarithm, i.e. $3^x = 3^2$, from which $x = 2$.

Hence **$\log_3 9 = 2$**

- (b) Let $x = \log_{10} 10$ then $10^x = 10$ from the definition of a logarithm, i.e. $10^x = 10^1$, from which $x = 1$.

Hence **$\log_{10} 10 = 1$** (which may be checked by a calculator)

- (c) Let $x = \log_{16} 8$ then $16^x = 8$, from the definition of a logarithm, i.e. $(2^4)^x = 2^3$, i.e. $2^{4x} = 2^3$ from the laws of indices, from which,

$$4x = 3 \text{ and } x = \frac{3}{4}$$

Hence **$\log_{16} 8 = \frac{3}{4}$**

Problem 2. Evaluate

- (a) $\lg 0.001$ (b) $\ln e$ (c) $\log_3 \frac{1}{81}$

- (a) Let $x = \lg 0.001 = \log_{10} 0.001$ then $10^x = 0.001$, i.e. $10^x = 10^{-3}$, from which $x = -3$

Hence **$\lg 0.001 = -3$** (which may be checked by a calculator).

- (b) Let $x = \ln e = \log_e e$ then $e^x = e$, i.e. $e^x = e^1$ from which $x = 1$

Hence **$\ln e = 1$** (which may be checked by a calculator).

- (c) Let $x = \log_3 \frac{1}{81}$ then $3^x = \frac{1}{81} = \frac{1}{3^4} = 3^{-4}$, from which $x = -4$

Hence **$\log_3 \frac{1}{81} = -4$**

Problem 3. Solve the following equations:

- (a) $\lg x = 3$ (b) $\log_2 x = 3$
(c) $\log_5 x = -2$

- (a) If $\lg x = 3$ then $\log_{10} x = 3$ and $x = 10^3$, i.e. **$x = 1000$**

- (b) If $\log_2 x = 3$ then **$x = 2^3 = 8$**

- (c) If $\log_5 x = -2$ then $x = 5^{-2} = \frac{1}{5^2} = \frac{1}{25}$

Problem 4. Write (a) $\log 30$ (b) $\log 450$ in terms of $\log 2$, $\log 3$ and $\log 5$ to any base

- (a) $\log 30 = \log(2 \times 15) = \log(2 \times 3 \times 5)$
 $= \log 2 + \log 3 + \log 5$

by the first law of logarithms

- (b) $\log 450 = \log(2 \times 225) = \log(2 \times 3 \times 75)$
 $= \log(2 \times 3 \times 3 \times 25)$

$$= \log(2 \times 3^2 \times 5^2)$$

$$= \log 2 + \log 3^2 + \log 5^2$$

by the first law of logarithms

$$\text{i.e. } \log 450 = \log 2 + 2 \log 3 + 2 \log 5$$

by the third law of logarithms

Problem 5. Write $\log \left(\frac{8 \times \sqrt[4]{5}}{81} \right)$ in terms of $\log 2$, $\log 3$ and $\log 5$ to any base

$$\log \left(\frac{8 \times \sqrt[4]{5}}{81} \right) = \log 8 + \log \sqrt[4]{5} - \log 81,$$

by the first and second laws of logarithms

$$= \log 2^3 + \log 5^{(1/4)} - \log 3^4$$

by the laws of indices

$$\text{i.e. } \log \left(\frac{8 \times \sqrt[4]{5}}{81} \right) = 3 \log 2 + \frac{1}{4} \log 5 - 4 \log 3$$

by the third law of logarithms.

Problem 6. Simplify

$$\log 64 - \log 128 + \log 32$$

$$64 = 2^6, 128 = 2^7 \text{ and } 32 = 2^5$$

$$\begin{aligned} \text{Hence } \log 64 - \log 128 + \log 32 \\ &= \log 2^6 - \log 2^7 + \log 2^5 \\ &= 6 \log 2 - 7 \log 2 + 5 \log 2 \\ &\quad \text{by the third law of logarithms} \\ &= 4 \log 2 \end{aligned}$$

Problem 7. Evaluate

$$\frac{\log 25 - \log 125 + \frac{1}{2} \log 625}{3 \log 5}$$

$$\begin{aligned} &\frac{\log 25 - \log 125 + \frac{1}{2} \log 625}{3 \log 5} \\ &= \frac{\log 5^2 - \log 5^3 + \frac{1}{2} \log 5^4}{3 \log 5} \\ &= \frac{2 \log 5 - 3 \log 5 + \frac{4}{2} \log 5}{3 \log 5} \\ &= \frac{1 \log 5}{3 \log 5} = \frac{1}{3} \end{aligned}$$

Problem 8. Solve the equation:

$$\log(x-1) + \log(x+1) = 2 \log(x+2)$$

$$\begin{aligned} \log(x-1) + \log(x+1) &= \log(x-1)(x+1) \\ &\quad \text{from the first} \\ &\quad \text{law of logarithms} \end{aligned}$$

$$= \log(x^2 - 1)$$

$$\begin{aligned} 2 \log(x+2) &= \log(x+2)^2 \\ &= \log(x^2 + 4x + 4) \end{aligned}$$

$$\text{Hence if } \log(x^2 - 1) = \log(x^2 + 4x + 4)$$

$$\text{then } x^2 - 1 = x^2 + 4x + 4$$

$$\text{i.e. } -1 = 4x + 4$$

$$\text{i.e. } -5 = 4x$$

$$\text{i.e. } x = -\frac{5}{4} \text{ or } -1\frac{1}{4}$$

Now try the following exercise

Exercise 44 Further problems on the laws of logarithms

In Problems 1 to 11, evaluate the given expression:

1. $\log_{10} 10\,000$ [4]
2. $\log_2 16$ [4]
3. $\log_5 125$ [3]
4. $\log_2 \frac{1}{8}$ [-3]
5. $\log_8 2$ $\left[\frac{1}{3}\right]$
6. $\log_7 343$ [3]
7. $\lg 100$ [2]
8. $\lg 0.01$ [-2]
9. $\log_4 8$ $\left[1\frac{1}{2}\right]$
10. $\log_{27} 3$ $\left[\frac{1}{3}\right]$
11. $\ln e^2$ [2]

In Problems 12 to 18 solve the equations:

12. $\log_{10} x = 4$ [10 000]
13. $\lg x = 5$ [100 000]
14. $\log_3 x = 2$ [9]
15. $\log_4 x = -2\frac{1}{2}$ $\left[\pm \frac{1}{32}\right]$
16. $\lg x = -2$ [0.01]
17. $\log_8 x = -\frac{4}{3}$ $\left[\frac{1}{16}\right]$
18. $\ln x = 3$ [e^3]

In Problems 19 to 22 write the given expressions in terms of $\log 2$, $\log 3$ and $\log 5$ to any base:

19. $\log 60$ [$2 \log 2 + \log 3 + \log 5$]
20. $\log 300$ $\left[2 \log 2 + \frac{1}{4} \log 5 - 3 \log 3\right]$
21. $\log \left(\frac{16 \times \sqrt[4]{5}}{27}\right)$ [$4 \log 2 - 3 \log 3 + 3 \log 5$]
22. $\log \left(\frac{125 \times \sqrt[4]{16}}{\sqrt[4]{81^3}}\right)$ [$\log 2 - 3 \log 3 + 3 \log 5$]

Simplify the expressions given in Problems 23 to 25:

$$23. \log 27 - \log 9 + \log 81 \quad [5 \log 3]$$

$$24. \log 64 + \log 32 - \log 128 \quad [4 \log 2]$$

$$25. \log 8 - \log 4 + \log 32 \quad [6 \log 2]$$

Evaluate the expressions given in Problems 26 and 27:

$$26. \frac{\frac{1}{2} \log 16 - \frac{1}{3} \log 8}{\log 4} \quad \left[\frac{1}{2} \right]$$

$$27. \frac{\log 9 - \log 3 + \frac{1}{2} \log 81}{2 \log 3} \quad \left[\frac{3}{2} \right]$$

Solve the equations given in Problems 28 to 30:

$$28. \log x^4 - \log x^3 = \log 5x - \log 2x \quad \left[x = 2\frac{1}{2} \right]$$

$$29. \log 2t^3 - \log t = \log 16 + \log t \quad [t = 8]$$

$$30. 2 \log b^2 - 3 \log b = \log 8b - \log 4b \quad [b = 2]$$

12.3 Indicial equations

The laws of logarithms may be used to solve certain equations involving powers—called **indicial equations**. For example, to solve, say, $3^x = 27$, logarithms to a base of 10 are taken of both sides,

$$\text{i.e.} \quad \log_{10} 3^x = \log_{10} 27$$

and $x \log_{10} 3 = \log_{10} 27$ by the third law of logarithms.

$$\begin{aligned} \text{Rearranging gives} \quad x &= \frac{\log_{10} 27}{\log_{10} 3} \\ &= \frac{1.43136\dots}{0.4771\dots} = 3 \end{aligned}$$

which may be readily checked.

(Note, $(\log 8 / \log 2)$ is **not** equal to $\lg(8/2)$)

Problem 9. Solve the equation $2^x = 3$, correct to 4 significant figures

Taking logarithms to base 10 of both sides of $2^x = 3$ gives:

$$\log_{10} 2^x = \log_{10} 3$$

$$\text{i.e.} \quad x \log_{10} 2 = \log_{10} 3$$

Rearranging gives:

$$\begin{aligned} x &= \frac{\log_{10} 3}{\log_{10} 2} = \frac{0.47712125\dots}{0.30102999\dots} = \mathbf{1.585} \\ &\quad \text{correct to 4 significant figures.} \end{aligned}$$

Problem 10. Solve the equation $2^{x+1} = 3^{2x-5}$ correct to 2 decimal places

Taking logarithms to base 10 of both sides gives:

$$\log_{10} 2^{x+1} = \log_{10} 3^{2x-5}$$

$$\text{i.e.} \quad (x+1) \log_{10} 2 = (2x-5) \log_{10} 3$$

$$x \log_{10} 2 + \log_{10} 2 = 2x \log_{10} 3 - 5 \log_{10} 3$$

$$x(0.3010) + (0.3010) = 2x(0.4771) - 5(0.4771)$$

$$\text{i.e.} \quad 0.3010x + 0.3010 = 0.9542x - 2.3855$$

$$\text{Hence} \quad 2.3855 + 0.3010 = 0.9542x - 0.3010x$$

$$2.6865 = 0.6532x$$

$$\text{from which} \quad x = \frac{2.6865}{0.6532} = \mathbf{4.11}$$

correct to 2 decimal places.

Problem 11. Solve the equation $x^{3.2} = 41.15$, correct to 4 significant figures

Taking logarithms to base 10 of both sides gives:

$$\log_{10} x^{3.2} = \log_{10} 41.15$$

$$3.2 \log_{10} x = \log_{10} 41.15$$

$$\text{Hence} \quad \log_{10} x = \frac{\log_{10} 41.15}{3.2} = 0.50449$$

Thus $x = \text{antilog } 0.50449 = 10^{0.50449} = \mathbf{3.195}$ correct to 4 significant figures.

Now try the following exercise

Exercise 45 Indicial equations

Solve the following indicial equations for x , each correct to 4 significant figures:

1. $3^x = 6.4$ [1.691]
2. $2^x = 9$ [3.170]
3. $2^{x-1} = 3^{2x-1}$ [0.2696]
4. $x^{1.5} = 14.91$ [6.058]
5. $25.28 = 4.2^x$ [2.251]
6. $4^{2x-1} = 5^{x+2}$ [3.959]
7. $x^{-0.25} = 0.792$ [2.542]
8. $0.027^x = 3.26$ [-0.3272]

12.4 Graphs of logarithmic functions

A graph of $y = \log_{10} x$ is shown in Fig. 12.1 and a graph of $y = \log_e x$ is shown in Fig. 12.2. Both are seen to be of similar shape; in fact, the same general shape occurs for a logarithm to any base.

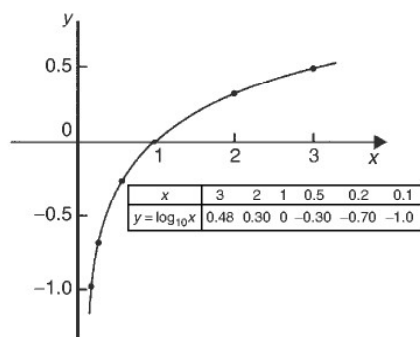


Figure 12.1

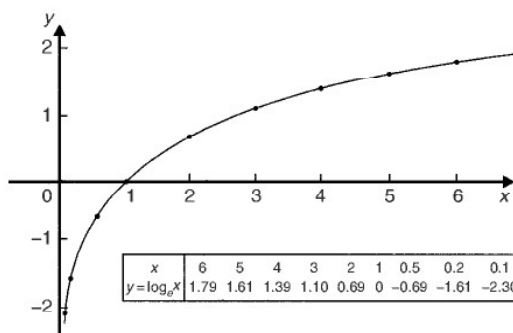


Figure 12.2

In general, with a logarithm to any base a , it is noted that:

(i) $\log_a 1 = 0$

Let $\log_a x = y$, then $a^y = x$ from the definition of the logarithm.

If $a^x = 1$ then $x = 0$ from the laws of logarithms.

Hence $\log_a 1 = 0$. In the above graphs it is seen that $\log_{10} 1 = 0$ and $\log_e 1 = 0$

(ii) $\log_a a = 1$

Let $\log_a a = x$ then $a^x = a$, from the definition of a logarithm.

If $a^x = a$ then $x = 1$

Hence $\log_a a = 1$. (Check with a calculator that $\log_{10} 10 = 1$ and $\log_e e = 1$)

(iii) $\log_a 0 \rightarrow -\infty$

Let $\log_a 0 = x$ then $a^x = 0$ from the definition of a logarithm.

If $a^x = 0$, and a is a positive real number, then x must approach minus infinity. (For example, check with a calculator, $2^{-2} = 0.25$, $2^{-20} = 9.54 \times 10^{-7}$, $2^{-200} = 6.22 \times 10^{-61}$, and so on.)

Hence $\log_a 0 \rightarrow -\infty$

Assignment 3

This assignment covers the material contained in Chapters 9 and 12. The marks for each question are shown in brackets at the end of each question.

1. Solve the following pairs of simultaneous equations:

(a) $7x - 3y = 23$

$2x + 4y = -8$

(b) $3a - 8 + \frac{b}{8} = 0$

$b + \frac{a}{2} = \frac{21}{4}$ (12)

2. In an engineering process two variables x and y are related by the equation $y = ax + \frac{b}{x}$ where a and b are constants. Evaluate a and b if $y = 15$ when $x = 1$ and $y = 13$ when $x = 3$ (4)

3. Transpose the following equations:

(a) $y = mx + c$ for m

(b) $x = \frac{2(y-z)}{t}$ for z

(c) $\frac{1}{R_T} = \frac{1}{R_A} + \frac{1}{R_B}$ for R_A

(d) $x^2 - y^2 = 3ab$ for y

(e) $K = \frac{p-q}{1+pq}$ for q (20)

4. The passage of sound waves through walls is governed by the equation:

$$v = \sqrt{\frac{K + \frac{4}{3}G}{\rho}}$$

Make the shear modulus G the subject of the formula. (4)

5. Solve the following equations by factorisation:

(a) $x^2 - 9 = 0$ (b) $2x^2 - 5x - 3 = 0$

(6)

6. Determine the quadratic equation in x whose roots are 1 and -3 (4)

7. Solve the equation $4x^2 - 9x + 3 = 0$ correct to 3 decimal places. (5)

8. The current i flowing through an electronic device is given by:

$$i = 0.005 v^2 + 0.014 v$$

where v is the voltage. Calculate the values of v when $i = 3 \times 10^{-3}$ (5)

9. Evaluate $\log_{16} 8$ (3)

10. Solve (a) $\log_3 x = -2$
(b) $\log 2x^2 + \log x = \log 32 - \log x$ (6)

11. Solve the following equations, each correct to 3 significant figures:

(a) $2^x = 5.5$

(b) $3^{2t-1} = 7^{t+2}$

(c) $3e^{2x} = 4.2$ (11)